Final assessment 2025-26

EC3040 Economics and Society

PART 1

Question 1:

Prompt given to ChatGPT : I´m going to send you a report of IFS on tax, private school fees and state school spending. Can you summarize the vat police based on the report and the summary should cover the main changes introduced by the policy and its likely costs and benefits. Also it should be 300 words max.

ChatGPT answer:

The policy analysed by the Institute for Fiscal Studies (IFS) involves **removing tax exemptions for private schools**, most importantly by **applying VAT at the standard 20% rate to private school fees**. After allowing for VAT deductions on inputs, exemptions for specialist provision, and VAT on boarding fees, the IFS estimates an **effective VAT rate of around 15%** on private school fees. The policy would also remove business rates relief, further increasing tax revenues

The main change introduced by the policy is a **rise in the post-tax cost of private schooling**, increasing average fees paid by parents. The IFS estimates that this package would raise around **£1.6 billion per year in additional tax revenue**. Importantly, the report argues that this revenue is unlikely to fall even if some parents withdraw from private schools, as any reduction in spending on fees would likely be reallocated to other goods and services that also attract VAT

On the spending side, higher private school fees may lead some pupils to move into the state sector. Based on existing evidence, the IFS judges demand for private schooling to be relatively inelastic, estimating a **3–7% fall in private school attendance** in the medium to long run. Accommodating these pupils would require additional public spending of roughly **£100–300 million per year**, depending on assumptions about marginal costs .

After accounting for these costs, the IFS concludes that the policy would likely generate a **net gain to the public finances of £1.3–1.5 billion per year**. This could fund around a **2% increase in state school spending in England**, which Labour proposes to target at disadvantaged pupils. However, the report highlights uncertainty around behavioural responses, tax avoidance, and long-run effects on inequality and educational outcomes .

Question 2:

The first claim that was generated was “Applying VAT at the standard 20% rate to private school fees.” On page 5 of the ifs report this claim is supported, however the summary leaves out what this new revenue would be used for. All numbers used in the chat gpt summary are correct, For example the claim that "effective VAT rate of around 15% on private school fees. The policy would also remove business rates relief, further increasing tax revenues” is on page 3 of the report.

The next claim is that the tax will raise 1.6billion GBP per year in revenue and even if some parents change it won't significantly change. This is related to what we learned in Lecture 2 about tax incidents where we learned that statutory incidence does not equal economic incidence because whoever pays the tax is not necessarily who has the costs. It is also seen in the IFS report on p.17 section 4. However, chat gpt stated that “ Importantly, the report argues that this revenue is unlikely to fall even if some parents withdraw from private schools, as any reduction in spending on fees would likely be reallocated to other goods and services that also attract VAT” this is supported in p.14 and p.19 but the wording does not convey the theory behind it.

The next claim is that demand for private school is relatively inelastic so most kids would stay in private school and only 3-7 % of pupils would move, so the government would need to spend an extra 100-300 million GBP annually. This claim is statistically supported by the IFS report (p.4) but could have gone further in detail.

Lastly the next claim that chat gpd produced was “net gain to the public finances of £1.3–1.5 billion per year. This could fund around a 2% increase in state school spending in England, which Labour proposes to target at disadvantaged pupils.” It is written as prediction when in the report it was multiple times stated that it's “reasonable to assume” it is support by lecture 2 slides and in the IFS report in pages 25- 27.

In conclusion the main statistical points are summarized correctly, and some background information is added to it. However, it relies too much on strong behavioral assumptions which could lead to information not being mentioned or wrongly misunderstood. It also did not mention anything about tax avoidance and reclassification risk and some points discussed in the lecture.

Chat gpt just assumes away key points such as its summary implying that the VAT revenue and enrolment responses are mechanical whereas in the lectures and the report emphasized that the behavioral response and tax incident are important to evaluating the policy.

Question 3:

The 2023 IFS report evaluates the economic impact of removing tax exemption from private schools in the United Kingdom and focuses its analysis on applying VAT to school fees of 20%. This policy is driven by social objectives and economic objectives. As it would raise revenues for state school and government (IFS 2023, p.4-5) , However its only goal is not financial the VAT revenue would be targeted at disadvantaged students (IFS, 2023, p.3) in order to address the wide resource gap between the two sectors as the average private school fees were "nearly 90% higher than state" school spending per pupil. (IFS, 2023, p.9). Another reason for this policy was promoting social mobility and social justice. They found evidence that proved that the higher earnings of private school graduates were more driven by network and sorting effects rather than the skills learned in school. (IFS, 2023, p.28)

Some of the economic rationale was that they estimated that this could generate around £1.6 billion in annual tax revenue for the government. However, there are concerns that this increase in private school fees will lead to a large increase in public school students. The report analyzes this concern by stating that even with the move from students it would still generate £1.3 -£1.5 billion since demand for private schools is relatively inelastic. Making the estimated pupil change of around 3-7 % which would be an additional spending of £100 million- £300 million on public schools. (IFS 2023, p)

Hence the report estimates that this extra revenue could fund a 2% boost to public school funding, which disadvantaged students would mainly benefit from. (IFS, 2023, p.27-28) However the report also mentions that there are uncertainties regarding tax avoidance (IFS, 2023, p.14-19) as

schools / parents might find ways to reduce their tax liability with things such as gifts or donations. Lastly the report also emphasizes that the private sector is not a monolith and that there is a high level of heterogeneity among schools (IFS,2023, page. 25) Therefore, the impact of the tax will vary based on the school's financial situation (Fee levels, School type, ability to absorb costs) .

PART 2

Question 1:

Q1

Benefit before paying fees

① $u_L = P_L + 10\sqrt{\frac{F}{y}}$

$u_M = P_M + 10\sqrt{F/y}$

$u_H = P_H + 10\sqrt{F/y}$

② $F = yt \rightarrow$ so $\frac{F}{y} = \frac{yt}{y} = t$

so the quality term becomes $10\sqrt{t}$

③ total private payment $\rightarrow \underset{\text{fee}}{t} + \overbrace{\tau t}^{\text{VAT on fee}} = t(1+\tau)$

④ Private school payoff = benefit − (fee + VAT) , $(t = \frac{F}{y})$

$$\pi_L^{priv} = P_L + 10\sqrt{t} - t(1+\tau)$$

$$\pi_M^{priv} = P_M + 10\sqrt{t} - t(1+\tau)$$

$$\pi_H^{priv} = P_H + 10\sqrt{t} - t(1+\tau)$$

The terms P_i $(Where = L, M, H)$ show heterogeneity in how much families value private schooling, it also reflects income and preferences where private schooling is mostly used by richer families as also seen in part 1. Then the term $10\sqrt{t}$ shows the benefit from resources and quality that is funded by the fees, so the higher the fees there will be more spending per pupil thus higher quality, which is the benefit from the private schools. $-t$ is the baseline cost of attending private schools (there does not need to be a VAT for this term to exist) then $-\tau t$ is the additional cost to the baseline form the VAT policy (increasing τ would make private schools less attractive). The VAT revenue would then be redistributed to public schools, which is not shown in the image above but is crucial later.

Question 2:

Scenario 1

↳ L chooses public

↳ M chooses private

↳ H chooses private

$x =$ # in public

$y = 3 - x \rightarrow$ # in private

$F = yt \rightarrow$ total private fee revenue

In scenario 1:

① $x = 1$, $y = 2$, $F = 2t$

② vat revenue $T \rightarrow T = yTt$

under $y=2$ we have $\Rightarrow T = 2Tt$

Payoffs

Public school payoff (for anyone in public)

$$u_i^{pub} = \frac{S}{x} + 5\frac{T}{x} \Rightarrow x = 1,\ T = 2Tt \Rightarrow u_{i(x=1)}^{pub} = S + 5(2Tt) = \underline{\underline{S + 10Tt}}$$

Private school payoff (for anyone in private)

$$u_i = p_i + 10\sqrt{F/y} \Rightarrow u_i^{priv} = u_i - t - Tt = p_i + 10\sqrt{\frac{F}{y}} - t(1+T)$$

But $\rightarrow \frac{F}{y} = \frac{yt}{y} = t \Rightarrow u_i^{priv} = p_i + 10\sqrt{t} - t(1+T)$

$$\left.\begin{aligned} u_M^{priv} &= p_M + 10\sqrt{t} - t(1+T) \\ u_H^{priv} &= p_H + 10\sqrt{t} - t(1+T) \end{aligned}\right\} \text{Payoffs}$$

No deviation inequalities

If L stays public $\rightarrow$ $S + 10\tau t$

If L goes priv then $y = 3$

$\hookrightarrow u_L^{priv} = P_L + 10\sqrt{t} - t(1+\tau)$

condition $\rightarrow$ $\boxed{S + 10\tau t \geq P_L + 10\sqrt{t} - t(1+\tau)}$ ← inequality

(L should not want to deviate)

M

if M stays private $\rightarrow$ $P_M + 10\sqrt{t} - t(1+\tau)$

if M $\rightarrow$ public then $x = 2$ and $y = 1$ so:

$T = 1 \cdot \tau t = \tau t$

$x = 2$

$$u^{Pub}_{(x=2)} = \frac{S}{2} + 5 \cdot \frac{\tau t}{2} = \frac{S + 5\tau t}{2}$$

$$\boxed{P_M + 10\sqrt{t} - t(1+\tau) \geq \frac{S + 5\tau t}{2}}$$ ← second inequality

H

if H deviates $x = 2$, $y = 1$, $T = \tau t$)

condition $\rightarrow$ $\boxed{P_H + 10\sqrt{t} - t(1+\tau) \geq \frac{S + 5\tau t}{2}}$

example parameter values:

$t = 100$

$\tau = 0.2$

$S = 0$

$P_L = 50$, $P_M = 100$, $P_H = 200$

i) $10\sqrt{t} = 10 \cdot 10 = 100$

$t(1+\tau) = 100(1.2) = 120$

ii) priv payoff

$u_L^{priv} = P_L + 100 - 120 = P_L - 20$

$u_L^{priv} = 30$

$u_M^{priv} = 80$

$u_H^{priv} = 180$

iii) Scenario 1 payoff for L

$S + 10\tau t = 0 + 10(0.2)(100) = 200$

Deviation public payoff $x = 2$

$\frac{S + 5\tau t}{2} = \frac{0 + 5(0.2)(100)}{2} = 50$

iii) L stays public $200 \geq 30$ ✓

M stays private $80 \geq 50$ ✓

H stays private $180 \geq 50$ ✓

Question 3:

Q3

Scenario 2 :

- L and M → Public school
- H → Private school
- $x = 2$
- $y = 1$

→ inequalities where no one wants to change + set of ** Satisfying the inequality.

Vat and Private school fees revenue :

$x = 2$, $y = 1$

Private fee revenue → $F = y \cdot t = 1 \cdot t = \underline{\underline{t}}$

VAT revenue

↳ each family pays → τt and $y = 1$ so

$T = y\tau t = \underline{\underline{\tau t}}$

Payoffs :

When $x = 2$ Public school payoff

Public school utility → $u_i^{Pub} = \frac{S}{x} + 5\frac{T}{x}$

$x = 2$, $T = \tau t$

↳ $u_{(x=2)}^{Pub} = \frac{S}{2} + 5 \cdot \frac{\tau t}{2} = \frac{S + 5\tau t}{2}$

$\underline{\underline{u_L^{Pub} = \frac{S + 5\tau t}{2}}}$, $\underline{\underline{u_M^{Pub} = \frac{S + 5\tau t}{2}}}$

Private payoff for H

$u_i^{Priv} = P_i + 10\sqrt{\frac{F}{y}} - t(1 + \tau)$

$\frac{F}{y} = \frac{yt}{y} = t$ so:

$u_i^{Priv} = P_i + 10\sqrt{t} - t(1 + \tau)$

$\underline{\underline{u_H^{Priv} = P_H + 10\sqrt{t} - t(1 + \tau)}}$

No deviation conditions

L stays public $\Rightarrow \frac{S+5\tau t}{2}$ L changes then $x=1$, $y=2$, $T=2\tau t$

Public Payoff $x=1$

$$u^{Pub}_{(x=1)} = S + ST = S + S(2\tau t) = S + 10\tau t$$

Private Payoff

$$u^{Priv}_{L} = PL + 10\sqrt{t} - t(1+\tau)$$

Condition $\Rightarrow$ $\boxed{\frac{S+5\tau t}{2} \geq P_L + 10\sqrt{t} - t(1+\tau)}$

(M)

same as L so $\boxed{\frac{S+5\tau t}{2} \geq P_M + 10\sqrt{t} - t(1+\tau)}$

(H)

H stays Priv $\Rightarrow P_H + 10\sqrt{t} - t(1+\tau)$

H changes Public $\Rightarrow x=3, y=0, T=0$

$$u^{Pub}_{(x=3)} = \frac{S}{3} + 5\cdot\frac{0}{3} = \frac{S}{3}$$

so $\Rightarrow$ $\boxed{P_H + 10\sqrt{t} - t(1+\tau) \geq \frac{S}{3}}$

Nash equilibrium

Numerical example

$t=100$, $\tau=0.2$, $S=600$, $P_L=0$, $P_M=20$, $P_H=200$

$10\sqrt{t} = 100$

$t(1+\tau) = 120$

Private Payoff $\Rightarrow u^{Priv}_{i} = P_i + 100 - 120 = P_i - 20$

$u^{Priv}_{L} = -20$

$u^{Priv}_{M} = 0$

$u^{Priv}_{H} = 180$

Public payoff

$$\frac{S+5\tau t}{2} = \frac{600 + 5(0.2)(100)}{2} = \frac{600+100}{2} = 350$$

if H deviates Public payoff:

$$\frac{S}{3} = 300$$

Inequalities:

L : $350 \geq -20$ ✓

M : $350 \geq 0$ ✓

H : $180 \geq 200$ × fails

need to increase P_H so $P_H = 240$

$u_H^{Priv} = 240 - 20 = 220$

H : $220 \geq 200$

all 3 conditions hold

Question 4:

If the term **S** (public school quality) increases in both scenarios it would make the public school more attractive. It is more likely in scenario 2 as it can tempt middle income households to switch to public and lower S makes scenario 1 more likely. Then the term **t** (Private school fee level) if t rises then $10\sqrt{t}$ increases slowly and $-t(1+\tau)$ would decrease linearly and faster. (higher t reduces private attractiveness) So higher t makes scenario 2 more likely as M will have to go public school, hence lower t makes scenario 1 more likely. The τ (VAT rate) would enter directly to private payoffs so if private cost would increase (by τt) it would push families towards the public school system. Making a direct negative impact on private by $-\tau t$ and would have a positive impact on public through the term T. So if it is higher it would make scenario 2 more likely and vice versa if it's lower it would make scenario 1 more likely.

Then we have the preference term such as $\boldsymbol{P_h}$. A higher Ph makes it easier for h to stay private which is important in scenario 2 because if H deviates all are in public. So, the higher P_h makes scenario 2 more feasible. Then P_m which if it increases it will directly increase middle class families' private payoff. Making scenario 1 more likely (M stays private) and a lower P_m makes scenario 2 also likely where M would join L in public school. The last preference term is P_l. Which increasing this would only affect Lower income families' private benefit. However, L in both scenarios is in public and the condition is whether they would deviate to private. Higher P_l makes L more likely to go private making the scenarios where L is public less stable. Making scenarios 1 and 2 harder to sustain and making it more difficult to have any equilibrium where L stays public.

Question 5:

1) Social welfare → $W = u_L + u_M + u_H$ Q5

2) VAT effects on utilities $u_i^{Pub} = \frac{s}{x} + 5\frac{y\tau t}{x}$

3) change in public utility when τ rises (by taking derivative):

$$\frac{\partial u_i^{Pub}}{\partial \tau} = 5\frac{yt}{x} > 0 \text{ if } \underbrace{y > 0}_{\text{(if anyone is in private)}}$$

anyone in private ($y>0$), increasing VAT raises the utility of each public school family.

4) Private school utility (taken from Q1)

$$u_i^{priv} = P_i + 10\sqrt{t} - t(1-\tau) \longrightarrow \frac{\partial u_i^{Priv}}{\partial \tau} = -t < 0$$

↑ vat leads to lower utility of each private school family by t per unit increase in τ.

5) there are x public families, each getting u^{Pub} and y private families each loosing t marginally from VAT.

$$\frac{\partial W}{\partial \tau} = x \cdot \frac{\partial u^{Pub}}{\partial \tau} + \sum_{\text{Private } i} \frac{\partial u_i^{Priv}}{\partial \tau}$$

we have know

$$x \cdot \frac{\partial u^{Pub}}{\partial \tau} = x \cdot 5\frac{yt}{x} = 5yt \quad \text{and} \quad \text{Private part} = y \cdot (-t) = -yt$$

$$\frac{\partial W}{\partial \tau} = 5yt - yt \Rightarrow \boxed{\frac{\partial W}{\partial \tau} = 4yt}$$

Based on calculations in the image above one can interpret that if Y > 0 then we would have the formula

$\frac{\partial W}{\partial \tau} = 4yt > 0$. Hence with fixed choices increasing VAT always increases total welfare.

If welfare decreases that would mean that an increase in τ that would result in a decrease in welfare is if it causes behavioral responses (so x,y changes)

Higher τ raises welfare holding all choices fixed, but it could reduce welfare if it changes the equilibrium so that large private school benefits are strongly negatively impacted or change in VAT base. To show this I compared welfare across scenarios 1 and 2 and looked at each family's utility changes. (can be seen in the image below)

Welfare across Scenarios:

Scenario 1 (L public, M and H private, $x = 1$, $y = 2$)

$W_1 = S + P_M + P_H + 20\sqrt{t} - 2t + 8\tau t$

Scenario 2 (L and M are public, H private, $x = 2$, $y = 1$)

$W_2 = S + P_H + 10\sqrt{t} - t + 4\tau t$

Welfare in all public ($x=3$, $y=0$)

$W_3 = S$

So $\uparrow \tau$ causes a move from scenario 1 to 2 welfare falls when:

$$\boxed{W_2 < W_1}$$

differences:

$W_1 - W_2 = P_M + 10\sqrt{t} - t + 4\tau t$ → So scenario 2 is worse then scenario 1 when: $\boxed{P_M + 10\sqrt{t} - t + 4\tau t > 0}$

How each family utility changes:

i) if family i is in private: $\frac{\partial U_i}{\partial \tau} = -t$

ii) if family i is in public (x=Public, y=private) : $\frac{\partial U_i}{\partial \tau} = 5\frac{yt}{x}$

taking values from scenarios:

1) ($x = 1$, $y = 2$)
→ L public : $10t$
→ M private : $-t$
→ H Private : $-t$

2) ($x = 2$, $y = 1$)
↳ L public : $\frac{5t}{2}$
↳ M Public = $\frac{5t}{2}$
↳ H Private = $-t$

From these results one can see that by holding school choices fixed, higher VAT will increase welfare if there is at least one family in the private school. Moreover, allowing the equilibrium to change, might lead to welfare falling if higher VAT makes families make the choice of leaving private schools despite the large private benefits.

Question 6:

Some of the benefits of introducing VAT to Private school fees are the following. Public revenue would increase, which then also goes into public schools. (5T/x) . There would be a redistribution towards lower-income families, who would benefit from increased public funding, and this would be done through taxation(week 2 slides) . There would also be potential welfare gains, by holding school choices fixed, increasing VAT raises total welfare when there is at least one family remaining in private school. Another benefit would equity efficiency trade-off, by
Taxing consumption of richer households can fund public goods that benefit lower income households.

However, there would also be some drawbacks, such as raising the cost by τt will directly decrease utility of the middle income households and high income households. Furthermore, as seen in question 2 and question 4 there is a possibility for taxes to distort behavior as raising VAT could lead to families from private schools making the change into public school. Also, Welfare falling could be because of a move from scenario 1 to 2 (or all families being in public school) destroying private benefits, which is the efficiency cost of taxation. It also raises the problem of crowding and peer effect if there are more kinds in public school which would increase x and therefore decrease S/x and T/x.

The main result from the VAT being introduced would be that public schools gain higher need funding as a result of the revenue gained from the tax and private schools would benefit less as it would decrease demand and would result in higher prices.

PART 3:

Question 1

Predictions:

1. Middle income families are more responsive to a VAT increase than high-income families
2. An increase in VAT raises public school funding per student, given that at least one family remains in private school.
3. An increase in VAT on private school fees increases public school enrolment.
4. The negative effect of VAT on private school attendance is stronger when private school fees (t) are high.

Question 2

For the first prediction, Middle- income families are more responsive to a VAT increase than high-income families, the variables that would be measured are school choice by income group and VAT rate. This prediction looks at heterogeneity. For income groups I would take data that shows household income bands, parental occupation and school choice so whether a child attends private or public school. Data sources would Table 3.1a Percentile points from 1 to 99 for total income before and after tax - GOV.UK , Household income - GOV.UK Ethnicity facts and figures , Distribution of parental occupation - Social Mobility Commission State of the Nation - GOV.UK and Schools, pupils and their characteristics, Academic year 2024/25 - Explore education statistics

For the second prediction, an increase in VAT raises public school funding per student, given that at least one family remains in private school. For this prediction the variables that could be measured are private school funding per pupil (total public school expenditure / number pupils) and vat revenue from private school. However VAT revenue would be difficult to observe at school level so I would use estimated data which is not ideal. Moreover data I would use is the IFS report for the estimates of VAT revenue from private school and Schools | Institute for Fiscal Studies.

The third prediction, an increase in VAT on private school fees increases public school enrolment. Variables measured for this prediction would be public school enrollment and looking at effectiveness of VAT rate applied to private fees. Public school enrollment is well measured and directly observed making it easier to test empirically. Once again the IFS report would be used for its how the VAT tax is introduced and its timing and School capacity, Academic year 2023/24 - Explore education statistics - GOV.UK

Lastly, the negative effect of VAT on private school attendance is stronger when private school fees (t) are high. For this prediction private school attendance, private school fees and vat policy would be assured. By looking at the number of pupils attending private school and average annual tuition fees by school or region and at the timing of introducing VAT. A data source that could be used it Private School Fees in the UK 2025 – Complete Guide to Costs and Admissions by EDVISION to measure fees and UK private school pupils 2025| Statista measures the number of students in private schools focusing on private day school and not boarding schools as there is no comparison for them in the public sector. Moreover the IFS report would also be used.

Question 3

The prediction chosen is " An increase in VAT on private school fees increases public school enrolment"

Regression equation :

$$PublicEnroll_{rt} = \alpha + \beta VATPolicy_{t} + \gamma_{r} + \delta_{t} + \varepsilon_{rt}$$

Meaning of terms:

$r \rightarrow$ regions

$t \rightarrow$years

$PublicEnroll_{rt} \rightarrow$ is the outcome variable. SO its the number of students enrolled in state-funded schools in region r at time t.

$\alpha \rightarrow$constant term

$VATPolicy_t \rightarrow$ this term explains whether the VAT policy is in place or not at time t. As in the prediction I need a variable that captures the moment VAT starts applying. Hence if the value is 0 before VAT excited and after VAT introduced the value will be 1

$\gamma_r \rightarrow$ region fixed effects, control for time differences across uk (things that differ across region but don't change over time for example a certain region might prefer private schools, or cultural attitudes to education, urban vs rural structure)
$\delta_t \rightarrow$ this is the year fixed effects.its a control for national trends affecting all areas (demographics, national education reforms)

$\varepsilon_{rt} \rightarrow$this is the error term , (shocks to the uk at t years.)

Key parameter of the equation is β as it measures the average change in public school enrolment once VAT is introduced. It answers the question of what happens to public school enrolment when VAT on private school fees is introduced? Because this term shows how much public school enrolment changes after VAT is introduced, comparing each region to itself over time, and without taking into account permanent regional differences and nationwide trends that do not relate to VAT.

My theory is that VAT makes private schooling more expensive which will result in private school students switching to public school.
My null hypothesis is $Ho : \beta = 0$ and my alternative hypothesis $H_1 : \beta \succ 0$.
To test whether VAT will increase public school enrolment.

Question 4

Endogeneity for my equation would mean that $VATPolicy_t$ is correlated to the error term meaning that β would not measure the true casual effect of VAT. Because of Behavioural responses (link to part 2) meaning that families choose school based on unobserved preferences, abilities and or values. As in the model in part 2 the preferences(Pl,Pm and Ph) affect school choice and are unobserved in the data. This causes endogeneity as regions with stronger preferences for privates schools may response differently to VAT and these unobserved preferences are in ε_{rt} and correlate with enrollment changes. So β captures pre existing trends not the VAT effect.

Another problem are omitted variables because public school enrolment depends on factors that are not fully controlled like quality and capacity. Which in the model of part 2 public utility depends on $S/x + 5T/x$ and S . This causes endogeneity because if public schools improve at the same time as VAT is introduced, β will overstate the effect of VAT and these omitted factors enter ε_{rt}. Lastly there is also policy endogeneity as VAT may be introduced because of trends in education. For example rising private school attendance or growing inequalities in education and pressure on public school funding. It causes endogeneity as $VATPolicy_t$ is not randomly assigned since it is introduced as a response to existing problems in education and inequality which also affects public school enrolment, and β does not reflect why VAT had to be introduced, not its effect.

A possible solution to these issues could be instrumental variables as it tries to isolate random variation in VAT exposure breaking the link between VAT and unobserved factors. It removes correlation between VAT and unobserved preferences. It also fixes omitted variables as lon as they do not affect the instrument if it is exogenous.

PART 4

There are five main claims made in both videos that the model in Part 2 and the empirical evidence in part 3 support/ help clarify. The negative claims are that "VAT on private school is unfair" , " It will push children into state schools " and " it will hurt middle- class families" , the more positive claims are that " It will improve public school funding", " It will hurt reduce inequalities ".

As seen in part 2 the model shows VAT increase the cost of private schooling (τt), which in questions 2 to 4 it can lead to middle income families having to go into the public school system which supports the claim that VAT affects enrolment and not only government revenue. However, as seen in the model low-income families benefit more if they use public school and middle income families are the marginal group so they might bear the cost. Whereas High- income families are the least responsive due to high P_h and inelastic demand for private schools. In both videos the arguments of fairness and efficiency were brought up. In Part2 question 5 one can see that VAT can raise welfare while holding choices fixed, but welfare might fall if behavioural responses are large.

Claims about enrolment changes can be tested empirically, for example whether public enrolment actually rises after VAT (prediction 2 in Part 3). Moreover, the claims about redistribution require evidence on which income will have switched schools and whether public funding per pupil increases. Different empirical strategies would help separate the real VAT effect from pre- existing trends.

There were some key arguments in both videos that were not brought up in the model one being the capacity constraints in public school. In the model it is assumed that the public schools can absorb as many students as possible with no difficulties whereas in the videos it was argued that they are already overcrowded. We could extend the model by allowing the public school utility to depend on x. Another Key argument that is not in the models but in the videos is the private school supply response. The first video talks about how they will cut fees or change structure and even in extreme cases close. As the VAT cost won't only fall on parents but also the schools. Our model

could adapt to this by allowing t to respond endogenously to VAT, in order to capture fee reductions or even closures. So that the fee depends on VAT and if it goes up then we would be able to observe the school choosing a new fee, so that private school fees are not fixed.

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